

1. Create a dataframe of ten rows, four columns with random values. Convert some values to nan values. Write a Pandas program which will highlight the nan values.

Code:

```
import pandas as pd
import numpy as np

df = pd.DataFrame(np.random.randn(10, 4),
                  columns=['A', 'B', 'C', 'D'])

df.iloc[1, 2] = np.nan
df.iloc[4, 0] = np.nan
df.iloc[7, 3] = np.nan

styled_df = df.style.highlight_null(color='yellow')

print(df)
styled_df
```

Output:

	A	B	C	D	E
0	1	1.32921	nan	-0.31628	-0.99081
1	2	-1.07082	-1.43871	0.564417	0.295722
2	3	-1.6264	0.219565	0.678805	1.88927
3	4	0.961538	0.104011	nan	0.850229
4	5	nan	1.05774	0.165562	0.515018
5	6	-1.33694	0.562861	1.39285	-0.063328
6	7	0.121668	1.2076	-0.00204021	1.6278
7	8	0.354493	1.03753	-0.385684	0.519818
8	9	1.68658	-1.32596	1.42898	-2.08935
9	10	-0.12982	0.631523	-0.586538	nan

2. Create a dataframe of ten rows, four columns with random values. Write a Pandas program to set dataframe background Color black and font color yellow.

Code:

```
import pandas as pd
import numpy as np

df = pd.DataFrame(np.random.randn(10, 4),
                  columns=['A', 'B', 'C', 'D'])

df.iloc[1, 2] = np.nan
df.iloc[4, 0] = np.nan
df.iloc[7, 3] = np.nan

styled_df = df.style.highlight_null(color='yellow')

print(df)
styled_df
```

Output:

	A	B	C	D
0	-0.602265	-0.786898	-1.400355	0.843817
1	-0.519526	-0.625030	NaN	0.056484
2	-1.325210	0.844345	0.266862	1.051179
3	0.575148	-0.675940	-0.476648	1.596903
4	NaN	0.544063	0.540031	0.311837
5	-2.586867	1.635806	-0.132548	1.180138
6	-0.585966	0.546293	-0.702978	-1.215418
7	-0.189841	-1.309411	0.091045	NaN
8	0.360531	0.439304	-0.770409	-0.688410
9	-1.321807	1.518584	0.713830	1.090538

3. Write a Pandas program to detect missing values of a given DataFrame. Display True or False.

Code:

```
import
pandas as pd
import
numpy as np

data = {
    'A': [1, 2, np.nan, 4],
    'B': [5, np.nan, 7, 8],
    'C': [9, 10, 11, np.nan]
}

df = pd.DataFrame(data)

print("Original
DataFrame:") print(df)

print("\nMissing Values
(True/False):") print(df.isnull())
```

Output:

Original DataFrame:

	A	B	C
0	1.0	5.0	9.0
1	2.0	NaN	10.0
2	NaN	7.0	11.0
3	4.0	8.0	NaN

Missing Values (True/False):

	A	B	C
0	False	False	False
1	False	True	False
2	True	False	False
3	False	False	True
